

AI in power optimisation: Key techniques

AI-enhanced design optimisation.

AI techniques are revolutionising the chip design process, enabling better power management before the chip even reaches production.

- *Machine learning for power prediction.* ML models analyse design parameters to predict power consumption at various workloads. These insights help engineers optimise designs for power efficiency without sacrificing performance.

- *Dynamic voltage and frequency scaling (DVFS).* AI algorithms dynamically adjust frequency and voltage levels to match workload demands, minimising power usage during idle periods.

- *Clock gating and power gating.* AI identifies underutilised cores or logic blocks and applies power gating (cutting power) or clock gating (pausing the clock signal) to reduce dynamic and leakage power.

For high-performance networking chips, these techniques ensure efficient packet handling, even at peak traffic loads.

Workload prediction and real-time power control. AI excels at analysing complex and variable workloads common in networking applications. By predicting workloads, AI can proactively optimise power usage:

- *Traffic pattern analysis.* AI algorithms use historical data and real-time inputs to forecast network traffic patterns. This enables the chip to prepare for high-demand periods while conserving energy during low traffic.

- *Adaptive power scaling.* Based on workload predictions, AI adjusts power levels across different chip components to minimise unnecessary energy consumption.

For instance, AI increases the power supplied to critical cores during peak traffic hours for efficient packet processing. Conversely, AI throttles these cores during lower workloads to save energy while

maintaining performance.

AI for thermal management. As transistors become smaller and denser, thermal hotspots pose a significant challenge for networking chips operating at high frequencies. Excessive heat can degrade performance, shorten chip lifespan, and cause failures. AI-driven thermal management techniques offer effective solutions:

- *Hotspot prediction and detection.* AI algorithms process sensor data and thermal simulations to detect hotspots before they escalate.

- *Dynamic thermal throttling.* AI dynamically adjusts voltage, frequency, or workload distribution across different chip regions to mitigate overheating.

By maintaining optimal thermal conditions, AI improves the energy efficiency and reliability of N3E-based networking chips, even during prolonged high-performance operations.

AI for power modelling and verification. Design verification is critical in ensuring power efficiency for chips in nanoscale nodes. Traditional methods are time-intensive and complex, but AI accelerates this process:

- *AI-assisted power simulations.* Machine learning models simulate power consumption across millions of workloads, providing accurate insights in less time.

- *Bottleneck detection.* AI identifies power-hungry areas or inefficient design components, enabling engineers to fine-tune the chip architecture iteratively.

This AI-enhanced verification ensures that power savings are achieved early in the design process, reducing time-to-market and improving overall performance.

Case study: Real-time AI optimisation

To understand the impact of AI, consider a high-performance networking chip designed for data centres with 400Gbps throughput:

Traditional power management.

Static optimisation methods result in power wastage during low-traffic periods or reduced throughput during peak loads.

AI-driven optimisation. AI dynamically adjusts voltage, frequency, and core utilisation based on real-time network traffic data. This results in 30% power savings with consistent throughput and lower latency.

AI-powered optimisation enables networking chips to meet fluctuating performance demands efficiently, addressing both power consumption and reliability challenges.

Future prospects and challenges

AI-driven power optimisation for N3E-based networking chips is a promising solution, but it also presents challenges:

Model complexity. AI models must process large volumes of design and runtime data, requiring robust algorithms and computational resources.

Energy overhead of AI. AI itself consumes power. Lightweight, low-power AI models are being developed for on-chip optimisation.

Scalability. Ensuring AI-driven power optimisation scales across different chip architectures and use cases is essential.

Future advancements may include edge AI processors embedded within chips, enabling autonomous power management without relying on external systems.

Integrating AI in power optimisation transforms the performance and energy efficiency of high-performance networking chips built on the N3E node. By enabling real-time workload prediction, adaptive scaling, and proactive thermal management, AI techniques address traditional methods' limitations and unlock nanoscale technology's full potential.

The combination of AI-driven innovation and advanced nodes like N3E ensures a sustainable future for data processing and communication technologies. **EFY**